

Venkatesh Puchakayala

Data Engineer

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SUMMARY

A highly skilled Data Engineer with 5 years of experience in developing and optimizing data pipelines, ETL processes, and real-time analytics. Proficient in Python, SQL, Java, Scala, and big data technologies like Hadoop, Spark, and Kafka. Experienced in cloud platforms such as AWS, Azure, and GCP. Adept at data visualization using Power BI and Tableau, with a strong background in machine learning model deployment and monitoring big data pipelines.

TECHNICAL SKILLS

Languages & Frameworks: Python, SQL, Java, Scala, Bash, Shell Scripting

Data Analysis & Visualization Tools: Power BI, Tableau

Big Data Technologies: Hadoop, Apache Spark, Apache Hive, Apache HBase, Apache Kafka, Apache Flink, Apache Pig, MapReduce, Apache Storm, Apache Cassandra

Cloud Platforms and Services: AWS, Azure, GCP, AWS Glue, AWS S3, AWS Lambda, AWS EMR, Azure Data Factory, Azure Data Lake Storage, Azure HDInsight, Azure Databricks, Azure Event Hubs, Azure Stream Analytics, Azure Kubernetes Service (AKS), Google Dataflow, Google Cloud Pub/Sub

Data Warehousing and Databases: Amazon Redshift, Azure Synapse Analytics, Google BigQuery, Snowflake, MySQL, PostgreSQL, Oracle, Microsoft SQL Server, MongoDB, Cassandra, Redis, DynamoDB, HDFS

Others: Docker, Kubernetes, Jenkins, Git, GitHub, GitLab, Ansible, Terraform, AWS CloudFormation, Azure DevOps, Helm, Prometheus, Grafana, Agile, Scrum, DevOps, DataOps, CI/CD

EXPERIENCE

Data Engineer, DXC Technology

Jan 2024 – present | Remote, USA

- Developed and optimized end-to-end data pipelines using Hadoop, Spark, Hive, and HBase, handling large volumes of data from multiple sources to support real-time and batch processing.
- Created complex MapReduce jobs and Hive queries to handle data transformations and aggregations, enhancing data processing capabilities.
- Developed streaming data pipelines using Apache Kafka, Apache Spark Streaming, and Azure Stream Analytics, enabling real-time analytics and anomaly detection for financial and e-commerce clients.
- Collaborated with data scientists to build and deploy machine learning models for predictive analytics using Azure Machine Learning, Spark MLlib, and Azure Databricks, improving business insights and decision-making.
- Designed and implemented ETL processes using Azure Data Factory and Apache Airflow, enabling real-time analytics for business teams and reducing data processing time by 40%.
- Utilized Azure Monitor, Apache Ambari, and custom dashboards to monitor and optimize the performance of big data pipelines and cloud resources, reducing costs and processing times.
- Developed complex SQL queries for analyzing customer behavior, leading to a 15% increase in targeted marketing effectiveness. Created stored procedures, views, and optimized queries for better performance.
- Conducted data profiling and anomaly detection to ensure data quality and improve data-driven decision-making.
- Built interactive dashboards and data visualizations using Power BI and Tableau to generate actionable business insights.

Senior Data Engineer, Gensole Infotech Pvt. Ltd.

Oct 2018 – Aug 2022 | Bangalore, India

- Engineered and optimized end-to-end ETL pipelines using Apache Spark, Python, and Apache Kafka to handle large volumes of data from multiple sources, ensuring seamless data ingestion, processing, and storage.
- Utilized AWS and GCP to build scalable and efficient data architectures, including data lakes, data warehouses, and real-time analytics platforms, ensuring high availability and performance.
- Deployed and managed data storage solutions using Amazon S3, and Google BigQuery optimizing data retrieval and analytics performance by designing efficient schemas and indexing strategies.
- Developed monitoring solutions using AWS CloudWatch and Databricks to track data flow, pipeline health, and system performance, optimizing data processing and reducing bottlenecks.
- Automated complex ETL workflows using AWS Kinesis, and AWS Glue, reducing data processing time by 40% and minimizing manual intervention for data validation and transformation.
- Modernized on-premises data warehouse infrastructure by migrating to cloud-based solutions using Snowflake, improving report generation speed by 60% and enhancing business intelligence capabilities.

DATA ANALYSIS PROJECTS

- Customer Segmentation Analysis:** Analyzed customer purchase data using clustering techniques in Python and SQL, improving marketing strategies by 20%.
- Sales Forecasting Model:** Built a time-series forecasting model in Power BI and SQL, increasing revenue accuracy predictions by 30%.
- Sentiment Analysis Dashboard:** Developed a real-time sentiment analysis tool using NLP and Power BI for social media insights.

EDUCATION

Lamar University

Master of Science; Major in Computer Science

August 2022 – May 2024

Texas, USA

Mahaveer Engineering College

Bachelor of Technology; Major in Electronics and Communication Engineering

August 2015 – May 2019

Hyderabad, India

CERTIFICATION

Microsoft Certified: Azure Data Engineer Associate